

Alma C. Castañeda-Leautaud

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Research Interests

- AI+ physics driven drug discovery
- LLMs and generative diffusion models
- Exploration of the accessible chemical space
- multi-data integration and scientific workflow automation
- Enhanced molecular simulations and free energy calculations
- Binding modes and molecular property predictions
- molecular representation as fingerprints

Education

Ph.D. Candidate in Chemistry

GPA: 3.75, UC San Diego, CA, USA, (2022–Present)

Thesis: *GNN-Driven Drug Design and Translocation Mechanism of SARS-CoV-2 NSP13*

PI: Prof. Rommie Amaro

M.Sc. Nanosciences

CICESE, Ensenada, B.C., México (2017–2020)

Thesis: *Proposal of a new antipsychotic free of cardio metabolic side effects by molecular dynamics*

PI: Prof. Sergio Águila Puentes

B.S. Nanotechnology

UNAM, Ensenada, B.C., México (2012–2017)

Thesis: *Disassembly and reassembly of an icosahedral virus*

PI: Prof. Rubén Darío Cadena Nava

Languages

Spanish (Native) and English (TOEFL IBT Score: 121, 2020)

Honors & Awards

- Fulbright-García-Robles Scholarship, USA (2022–2025)
- CONACYT Scholarship, France (2020)
- CONACYT Scholarship, México (2017–2019)
- PRONABES scholarship, Mexico (2012–2016)

Work & Research Experience

- *Research Intern* at **Iambic Therapeutics, Inc.**, USA (2026)
- *Research Intern* at **Midwest-AviDD** Center, USA (2023–)
- *Biotesting Lab Coordinator* at Quality Control Department, **Mettler-Toledo**, México (2021–2022)

Conferences

- Gordon Research Conferences: Computational chemistry, Barcelona, Spain (2026), poster
- Drug-Discovery Consortium symposium, Riverside, USA (2025), poster
- ACS Spring National Meeting, San Diego, USA, (2025), poster
- Gulf Coast Undergraduate Research Symposium, Rice Univ. (2021), speaker
- III Coloquio de Simulaciones Computacionales (2020), speaker
- Symposium of Nanoscience and Nanomaterials (2020, 2019), speaker/poster

Publications

1. Castañeda-Leautaud, A. C., López-Pérez, K., Chung, D., Bannister, T. D., Spicer, T. P., Harris, R. S., Scampavia, L., Miranda-Quintana, R. A., & Amaro, R. E. (2026). *An HTS-derived AI evaluation dataset with realistic virtual screening space and non-trivial class separation*, **NeurIPS-2026** conference (under review).
2. Castañeda-Leautaud, A. C., Bannister, T. D., Kim, E., Chung, D., & Amaro, R. E. (2026). *Discovery of Two Novel Small-Molecule Series with Potent SARS-CoV-2 Inhibition via Putative nsp13 Open-State Trapping*, **RSC Chemical Science** (under review).
3. Castañeda Leautaud, A. C., & Amaro, R. (2025). *Optimal message passing for molecular prediction is simple, attentive, and spatial*, **RSC Digital Discovery**. doi.org/10.1039/D5DD00193E.
4. Castañeda Leautaud, A. C., Srivastava, A., Eun Jung, K., Chung, D., Bannister, T. D., & Amaro, R. E. (2025). *Discovery of tetrahydroisoquinoline-based SARS-CoV-2 helicase inhibitors with iterative, deep learning enhanced virtual screening*, **J. of Chem. Inf. and Model.** doi.org/10.1021/acs.jcim.5c02059.
5. Castañeda-Leautaud, A. C., Vidal-Limón, A., & Águila, S. A. (2022). *Molecular dynamics and free energy calculations of clozapine bound to D2 and H1 receptors*, **J. of Biom. Struct. and Dynam.**, 41(19), 9313–9325. doi.org/10.1080/07391102.2022.2148748

Technical Skills

Programming: Python, Bash, csh

Python Libraries: PyTorch, PyTorch Geometric, NetworkX, NumPy, CuPy, pandas, matplotlib, scikit-learn, RDKit, API requests, asyncio

Artificial Intelligence: Self-supervised learning (transformer-based protein language models), supervised learning (GNNs, MPNNs, CNNs, MLPs, Random Forest, SVM, Naive Bayes, XGBoost), and unsupervised learning (PCA, UMAP, KNN, t-SNE)

Biomolecular Modeling and Simulations: AMBER, NAMD, VMD, CHARMM-GUI, AutoDock, Schrödinger, MDAnalysis, ProDy, Open Babel, Gaussian, Spartan

Cloud and Data Infrastructure: AWS, REST APIs, JSON data pipelines, scientific database integration (PDB, DrugBank, OpenTargets, PubChem, UniProt), biomedical database integration

Courses & Training

Programming & Data Science

- Python Data Structures (UMich, 2021)
- Accessing Web/Network Data (UMich, 2021)
- Advanced Python (DataCamp, 2020)
- Python 101 for Data Science (IBM, 2020)
- Applied Data Science with Python (IBM, 2020)
- Data Mining (III Coloquio en Simulaciones Computacionales, 2020)

Computational & Physical Sciences

- Molecular Dynamics (UNAM, 2018)
- Ferromagnetic Fluids (Clubes de Ciencia, 2015)
- Topological Phases (Clubes de Ciencia, 2014)